DATA SET – 3

Business Insights Report

Northwind Traders

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INTRODUCTION

Northwind Traders Dataset:

Northwind Traders is a sample business dataset used to study sales, customers, products and employees. It contains details about orders, products, customers and shipping.

Purpose of the Dataset:

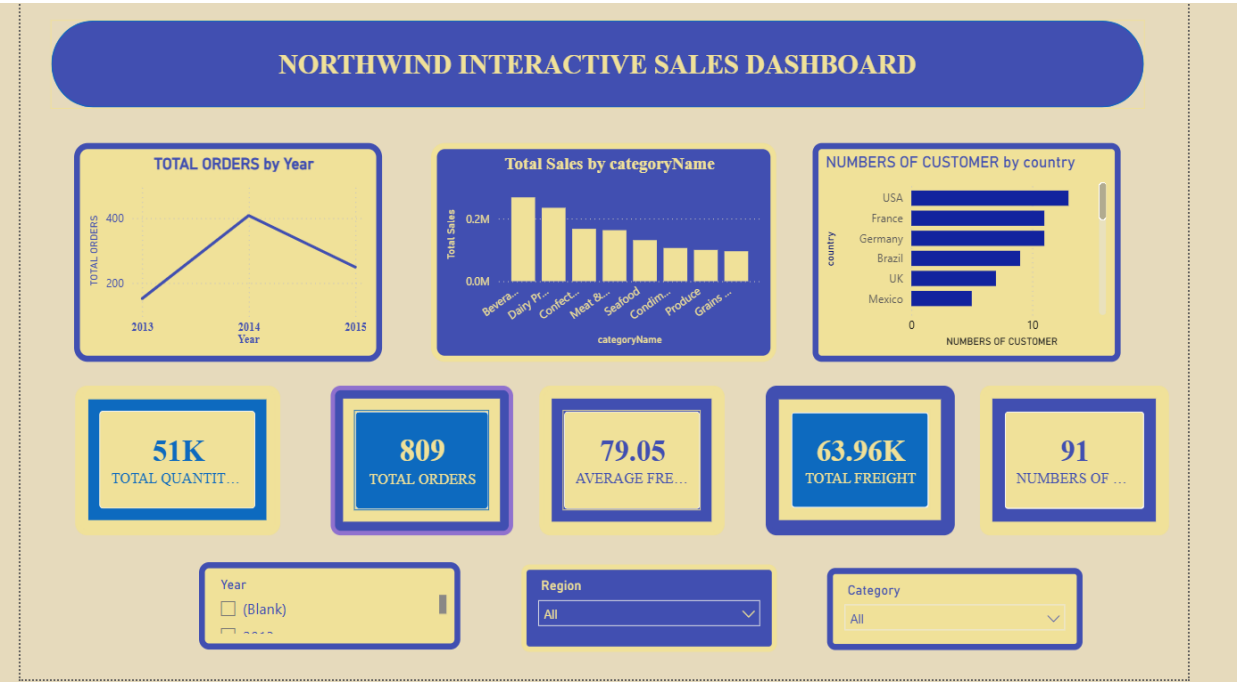
The dataset helps to understand business performance, sales growth, customer demand and product performance.

Purpose of the Business Insights Report:

The purpose of this report is to analyse business performance using Power BI. It helps to understand sales, orders, customers and employee activities.

DASHBOARD

Dashboard Screenshot:



Purpose of the Dashboard:

The dashboard shows important business information in a simple visual format. It helps to understand sales, orders, customers and employee performance.

Key Components:

1. Total Orders by Year: Shows the total number of orders received each year.
2. Total Sales by Product Category: Displays sales amounts for different product categories.
3. Customers by Country: Shows the number of customers in each country.
4. Total Quantity Sold: Displays the total number of products sold.
5. Total Orders: Shows the total number of orders received.
6. Average Freight: Displays the average delivery charges per order.
7. Total Freight: Shows the total transportation charges.
8. Number of Customers: Displays the total number of customers.
9. Total Orders by Employee: Shows the number of orders handled by each employee.

Interactive Filters Available:

- **Year**: Shows business performance for different years.
- **Region**: Helps compare different regions.
- **Category**: Shows sales details of different product categories.

These filters help users view the required information easily.

Purpose of Slicers:

The purpose of slicers is to filter and view specific data easily. They help users compare business performance and analyse information based on their selected requirements.

KPI ANALYSIS



Explanation of KPIs:

1. Total Orders

The company received a total of 809 orders during the selected period.

2. Total Quantity Sold

A total of 51,317 products were sold to customers.

3. Total Freight

The total freight amount is approximately 63.96K. It represents the transportation charges.

4. Average Freight

The average freight amount is 79.05 per order.

5. Number of Customers

The company has 91 customers in the dataset.

Key Performance Indicators:

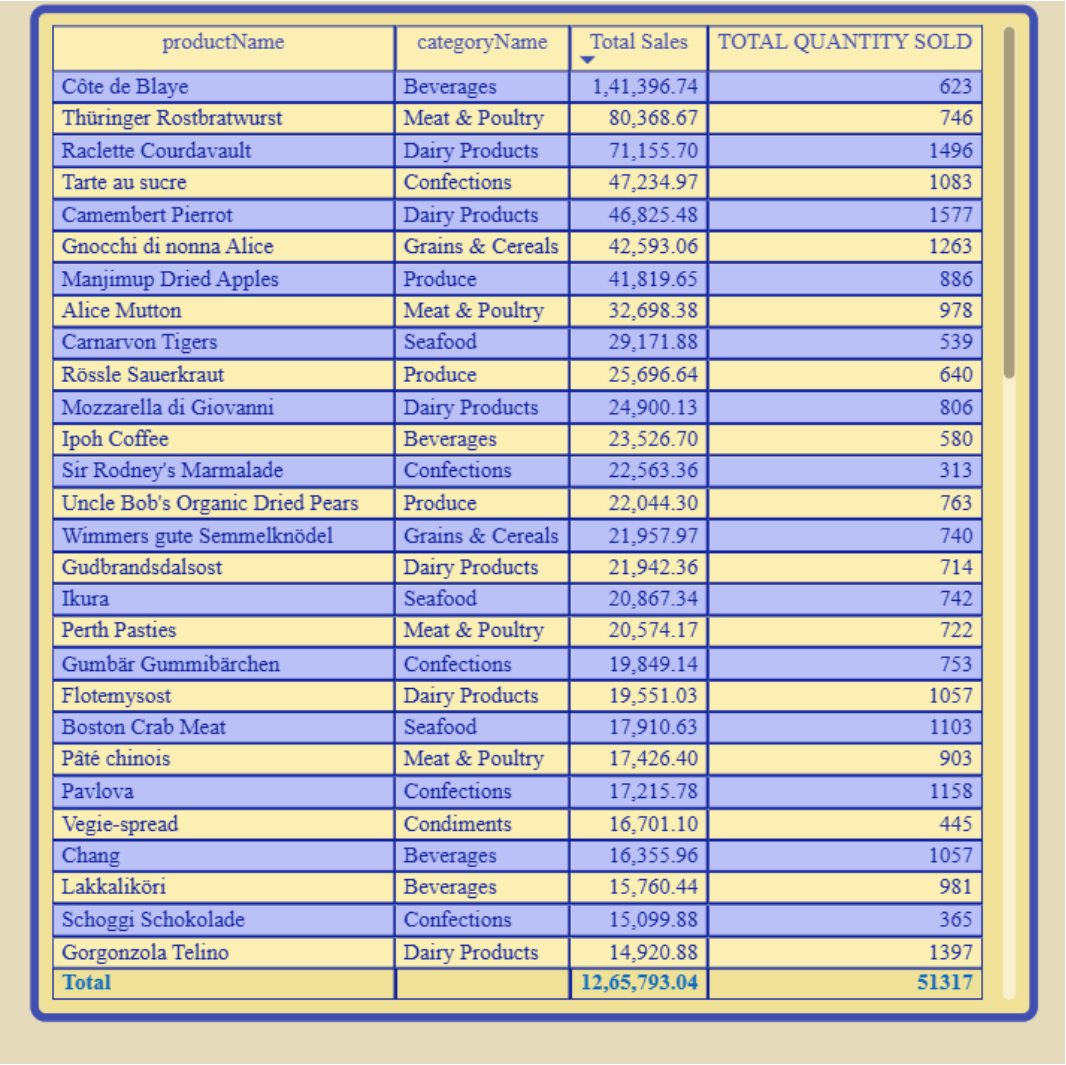
Particulars	Values
Total Orders	809
Total Quantity Sold	51,317
Total Freight	63.96K
Average Freight	79.05
Number of Customers	91

PRODUCT AND SALES DETAILS

Product and Customer Analysis:

This analysis shows the sales performance of different products and categories. It helps identify which products have higher and lower sales. It provides information about total sales and quantity sold in each category. It also helps compare product demand and understand customer preferences. This analysis helps the company identify popular products and improve the sales of low-performing products.

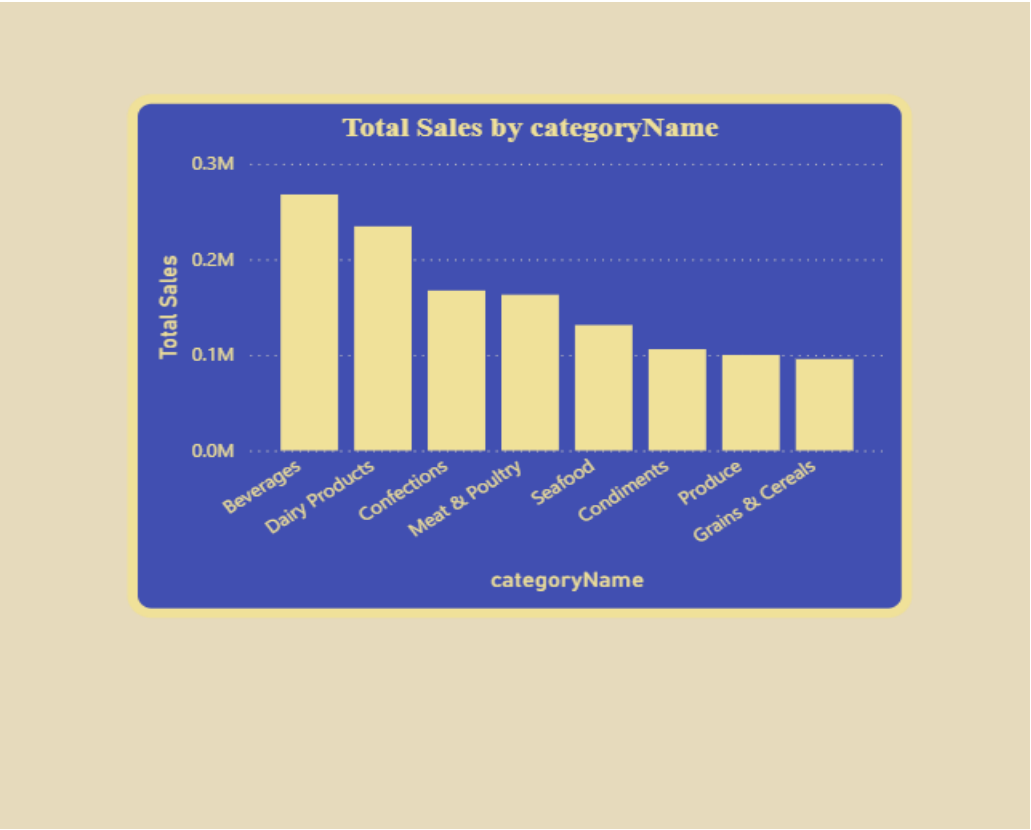
- Total Sales: ₹12,65,793.04
- Total Quantity Sold: 51,317 products.



productName	categoryName	Total Sales	TOTAL QUANTITY SOLD
Côte de Blaye	Beverages	1,41,396.74	623
Thüringer Rostbratwurst	Meat & Poultry	80,368.67	746
Raclette Courdavault	Dairy Products	71,155.70	1496
Tarte au sucre	Confections	47,234.97	1083
Camembert Pierrot	Dairy Products	46,825.48	1577
Gnocchi di nonna Alice	Grains & Cereals	42,593.06	1263
Manjimup Dried Apples	Produce	41,819.65	886
Alice Mutton	Meat & Poultry	32,698.38	978
Carnarvon Tigers	Seafood	29,171.88	539
Rössle Sauerkraut	Produce	25,696.64	640
Mozzarella di Giovanni	Dairy Products	24,900.13	806
Ipoh Coffee	Beverages	23,526.70	580
Sir Rodney's Marmalade	Confections	22,563.36	313
Uncle Bob's Organic Dried Pears	Produce	22,044.30	763
Wimmers gute Semmelknödel	Grains & Cereals	21,957.97	740
Gudbrandsdalsost	Dairy Products	21,942.36	714
Ikura	Seafood	20,867.34	742
Perth Pasties	Meat & Poultry	20,574.17	722
Gumbär Gummibärchen	Confections	19,849.14	753
Flotemysost	Dairy Products	19,551.03	1057
Boston Crab Meat	Seafood	17,910.63	1103
Pâté chinois	Meat & Poultry	17,426.40	903
Pavlova	Confections	17,215.78	1158
Vegie-spread	Condiments	16,701.10	445
Chang	Beverages	16,355.96	1057
Lakkaliköri	Beverages	15,760.44	981
Schoggi Schokolade	Confections	15,099.88	365
Gorgonzola Telino	Dairy Products	14,920.88	1397
Total		**12,65,793.04**	**51317**

Key Details

- Total Sales: ₹12,65,793.04
- Total Quantity Sold: 51,317 products.
- The dashboard displays eight product categories.
- It helps compare the sales performance of different categories.



Product Category Performance:

Particulars	Observation
Highest Selling Category	Beverages
Lowest Selling Category	Grains & Cereals

Business Observation

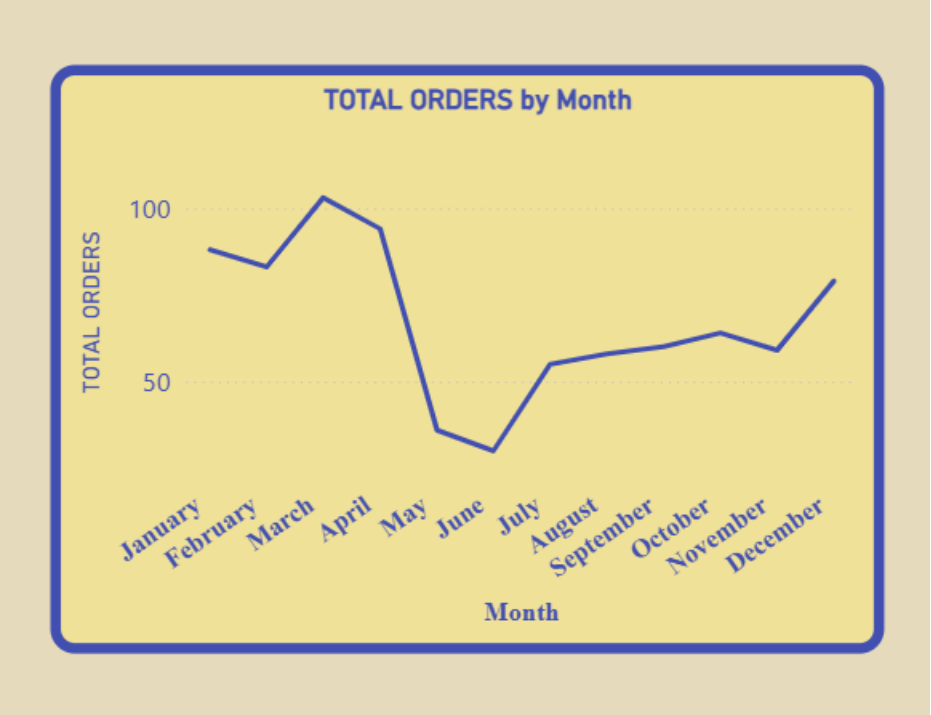
Beverages recorded the highest sales, while Grains & Cereals recorded the lowest sales. The company should maintain popular products and improve the sales of low-performing categories.

Findings

1. Beverages is the best-performing category.
2. Grains & Cereals has the lowest sales.

MONTHLY ORDER TREND ANALYSIS

The monthly order trend shows the number of orders received during different months of the year.



Analysis

1. January and February recorded good order activity.
2. March recorded the highest number of orders.
3. Orders decreased during April.
4. May and June showed a further decrease.
5. July recorded the lowest orders.
6. Orders gradually increased from August to December.

Business Observation

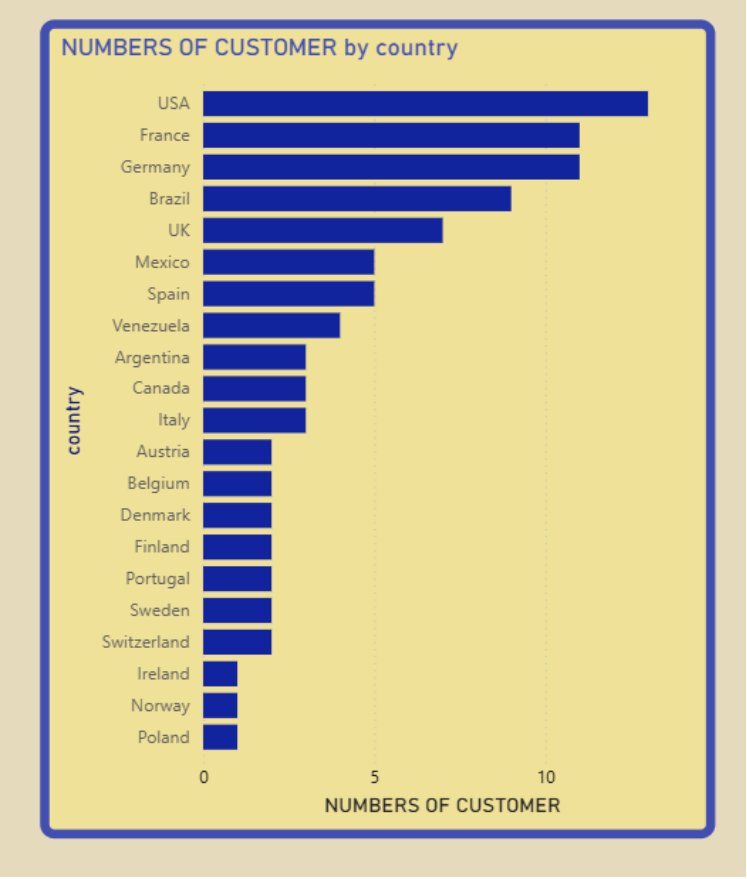
The company received more orders in March and fewer orders in July. Promotional offers can help increase orders during low-performing months.

Findings

Particulars	Findings
Highest Order Month	March
Lowest Order Month	July

REGIONAL ANALYSIS

Regional analysis shows the customer distribution across different countries.



Customer Distribution:

The company has customers from countries such as:

USA, France, Germany, Brazil, UK, Mexico, Spain, Venezuela, Argentina, Canada, Italy, Austria, Belgium, Denmark, Finland, Portugal, Sweden, Switzerland, Ireland, Norway and Poland.

Key Insights:

1. USA has the highest number of customers.
2. France and Germany also have more customers.
3. Brazil contributes to the customer base.
4. UK and Mexico also have good customer participation.
5. The company serves customers from many countries.

Regional Customer Details:

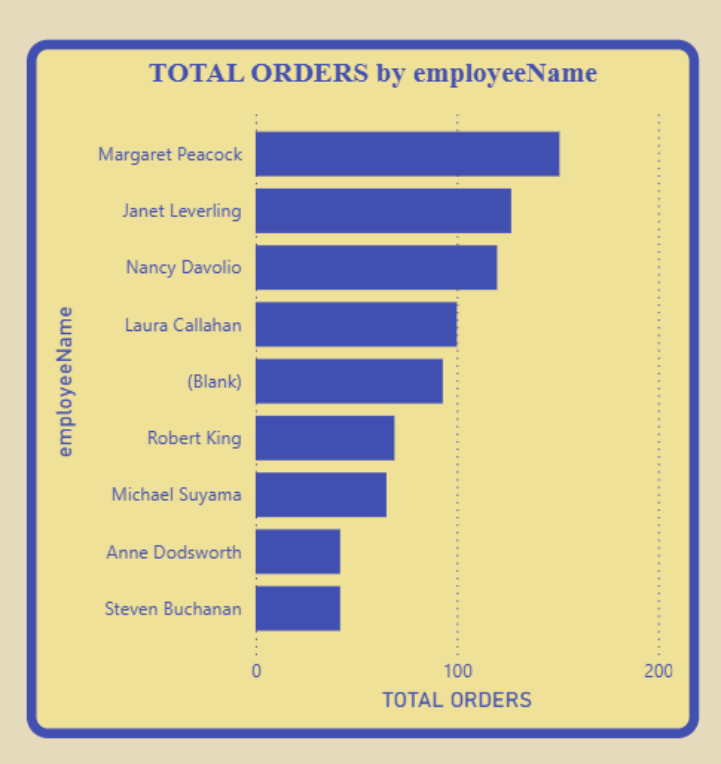
Particulars	Findings
Highest Customer Region	USA
Second Highest	France and Germany
Lowest Customer Regions	Ireland, Norway and Poland
Overall Distribution	Customers are located in different countries.

Business Observation:

USA has the highest customer count. The company should attract more customers from countries with fewer customers.

EMPLOYEE PERFORMANCE ANALYSIS:

Employee performance analysis shows the number of orders handled by different employees.



Key Insights

1. Margaret Peacock recorded the highest number of orders.
2. Janet Leverling recorded the second-highest performance.
3. Nancy Davolio also contributed to order activities.
4. Laura Callahan contributed to order management.
5. The chart helps compare employee performance.

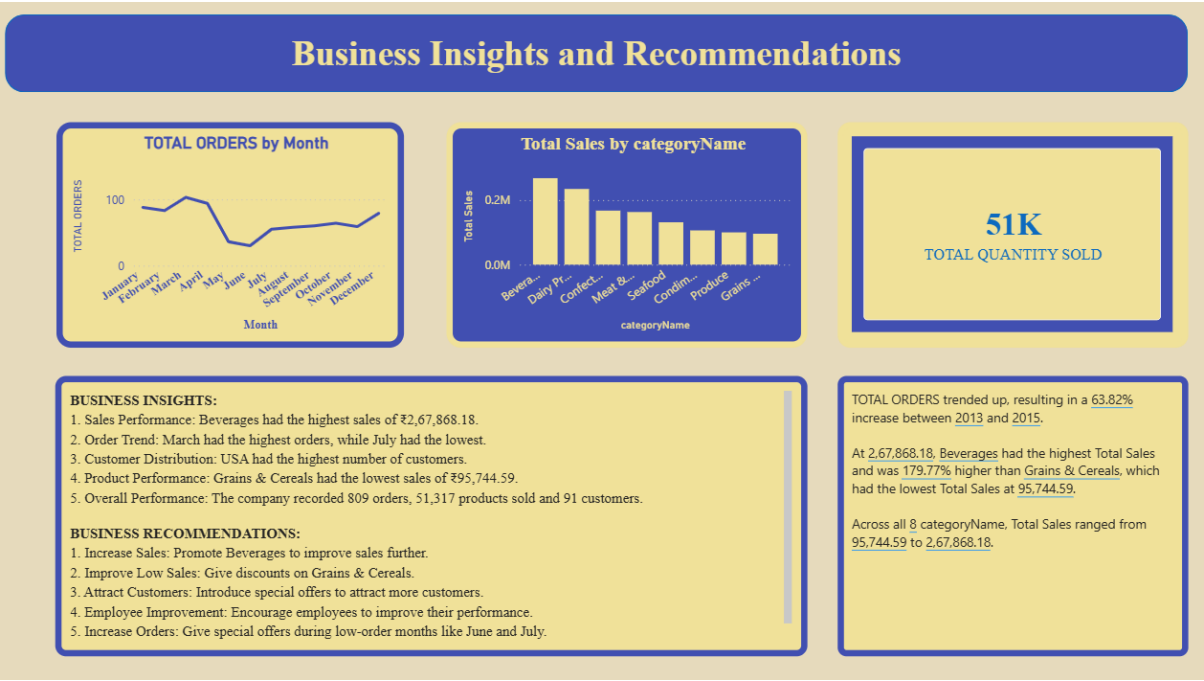
Employee Performance DetailsL

Particulars	Findings
Top Performing Employee	Margaret Peacock
Second Performing Employee	Janet Leverling
Third Performing Employee	Nancy Davolio

Business Observation

Margaret Peacock recorded the highest number of orders. The company should encourage all employees to improve their performance.

BUSINESS INSIGHTS



1. Sales Performance: Beverages recorded the highest sales among all categories.

2. Order Trend: March had the highest orders, while July had the lowest orders.

3. Customer Distribution: USA has the highest number of customers.

4. Product Performance: Grains & Cereals recorded the lowest sales.

5. Employee Performance: Margaret Peacock recorded the highest number of orders.

6. Overall Performance: The company recorded 809 orders and sold 51,317 products to 91 customers.

BUSINESS RECOMMENDATIONS

1. Improve Sales: Provide special offers to increase product sales.

2. Maintain Stock: Keep sufficient stock of high-selling products like Beverages.

3. Customer Growth: Attract new customers through advertisements and offers.

4. Employee Performance: Encourage employees and provide proper training.

5. Monthly Orders: Introduce discounts during low-order months like June and July.

6. Expand Business: Focus on countries with fewer customers to increase sales.

CONCLUSION

This Business Insights Report helped to understand the business performance of Northwind Traders using Power BI. The dashboard provided information about sales, orders, products, customers and employees. The analysis showed that Beverages had the highest sales, USA had the highest number of customers, March recorded the highest orders and Margaret Peacock

performed well in order management. Overall, Power BI helped to analyse business data, identify important trends and make better business decisions.